Plannr

Timetable, attendance and scheduling for SICSR

Full presentation

GROUP MEMBERS

Paras Deshpande

Sonal Mohrir

Tanisha Nath

COURSE

BCA Sem VII, Div A

Web UI and Content Management, Assessment 3

Faculty: Shashank T. Gaikwad

plannr.paraswtf.com

Agenda

01 The problem

02 Our research

03 What we found

04 How we prioritised

05 Plannr's features

06 Technology and system

07 Testing and demo

08 Conclusion

The timetable is hard to use

- time-table.sicsr.ac.in is an MRBS install: PHP + MySQL, a LAMP application
- One page for the whole college, built for a desktop, over plain HTTP
- Its day view is a single 141 KB HTML document listing every room
- Coordinators book each lecture by hand in the "Add Entry" form
- Information is split across the timetable site, Gmail, WhatsApp and Moodle

73%

say it doesn't tell them when a class changes

82%

say finding their own classes takes too many steps

82%

can't see their week at a glance

87%

use WhatsApp to learn about changes

Four places for one question

Timetable site
Classes and rooms

Gmail
Every change by hand

WhatsApp
The news second-hand

Moodle
Assignments, tests, attendance

WHAT IT COSTS

"Once, the class mentioned in the mail was different from the one in the timetable, and I only got to know about it later with my friend."

Student, SICSR

"Many times classes change with no email about it or cancel at the wrong moment"

Student, SICSR

How we researched

Group	Owner	Method
Students	Sonal Mohrir	Four open questions, a survey
Faculty	Tanisha Nath	Interviews, in person and by phone
Coordinators	Tanisha Nath	Interviews, observing their screens
System admins	Paras Deshpande	Not yet interviewed

12 to 23 September 2026 · Open questions · A survey · Interviews in person and by phone · Observing coordinators' screens

Who we heard from

15

Students

Answered four open questions

11

Students surveyed

Years 3 and 4 (9 in year 4, 2 in year 3)

10

Faculty

5 at SICSR, 5 at SIG; 6 in person, 4 by phone

4

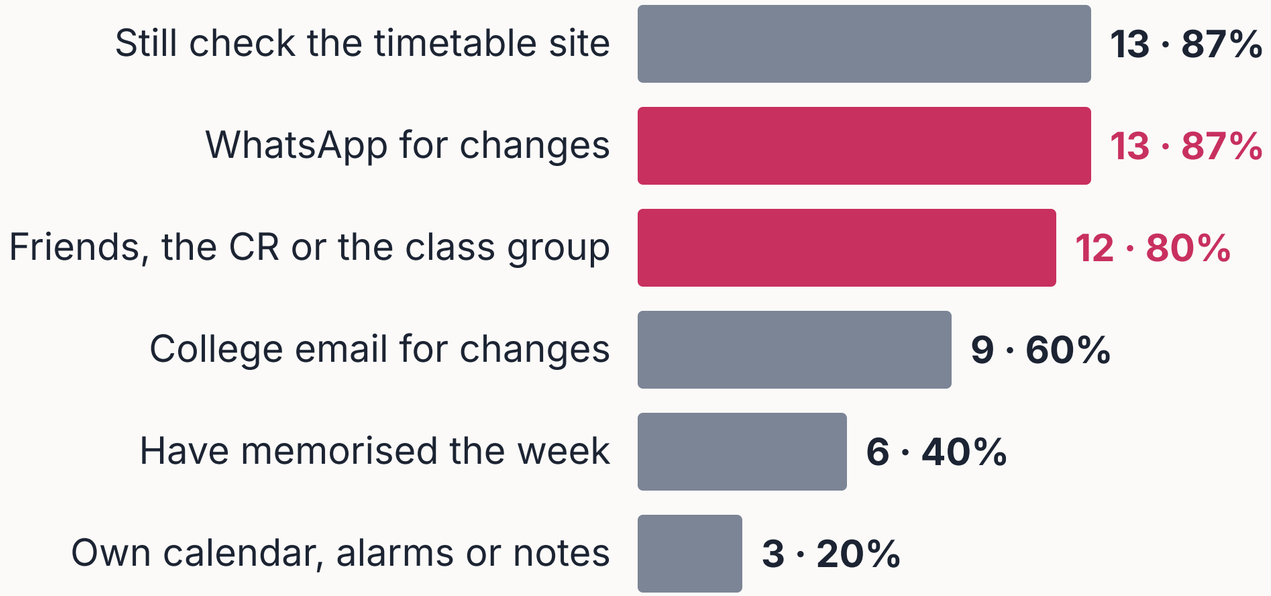
Coordinators

SICSR, interviewed in person

System admins: not yet interviewed.

How students keep up

HOW THEY FOLLOW THEIR SCHEDULE · n = 15 · TICK ALL



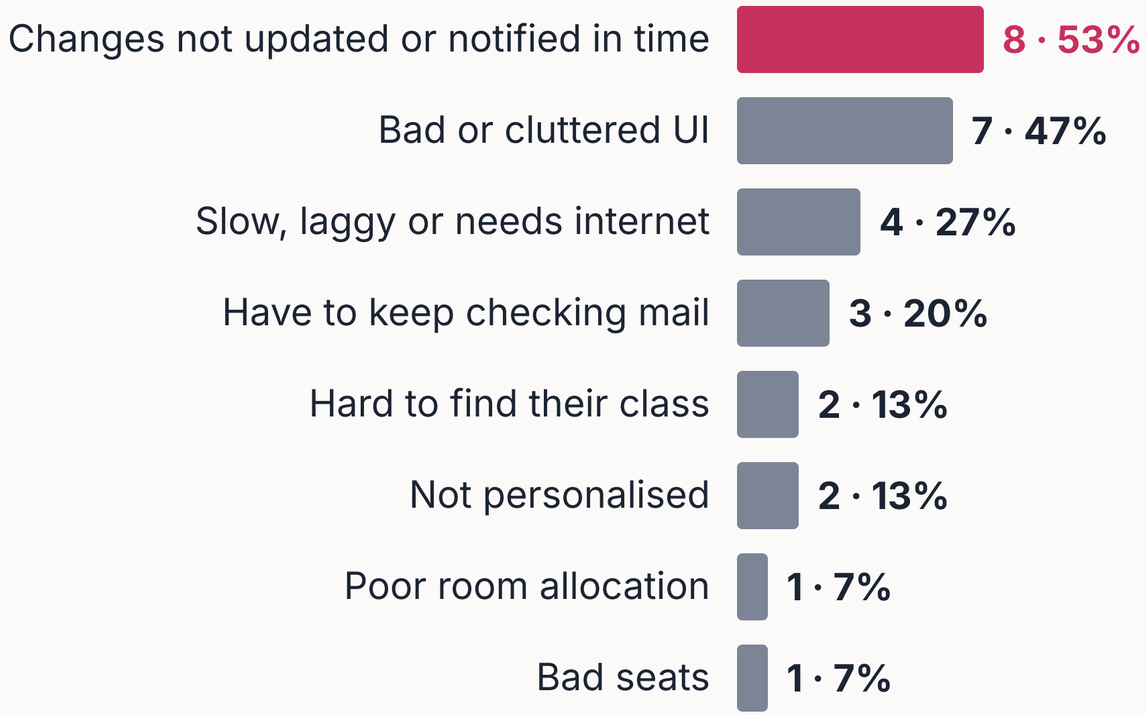
12 of 15

confirm with a friend, the CR or the group

The site is still open, but the news travels on WhatsApp.

What students raised unprompted

PROBLEMS RAISED IN THEIR OWN WORDS · n = 15



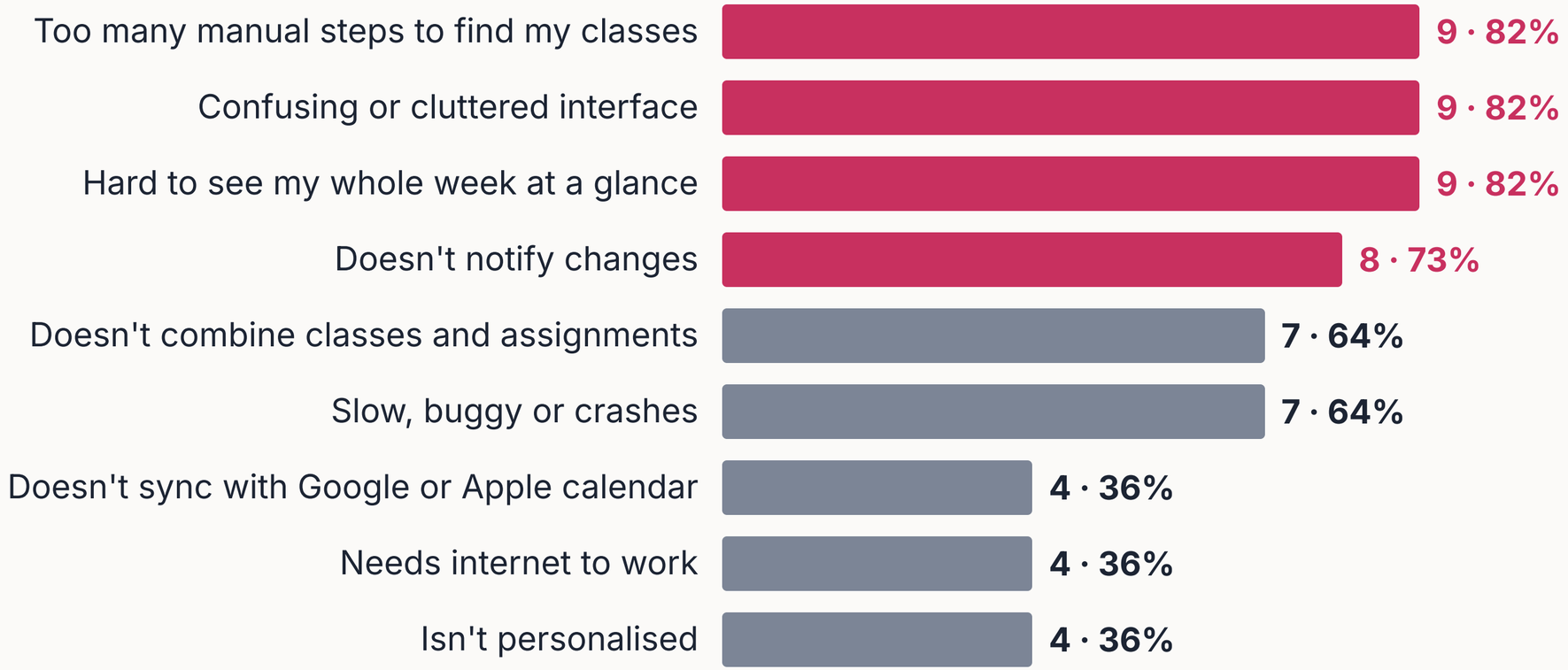
11 of 15

reported at least one problem without being prompted

Missed or late change notices came up more than any other issue.

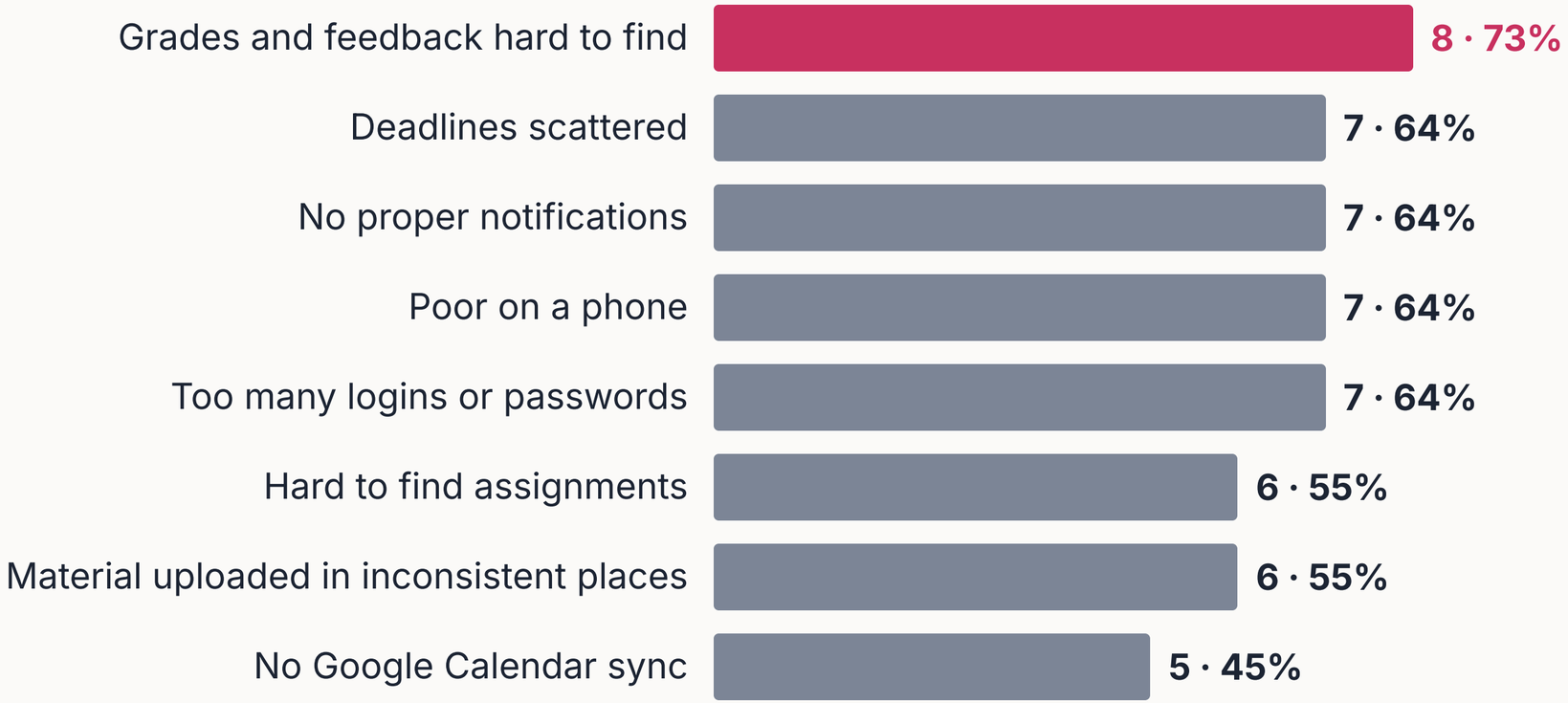
Problems with the timetable site

TICK ALL THAT APPLY · n = 11 · HIGHLIGHT = 73% OR MORE



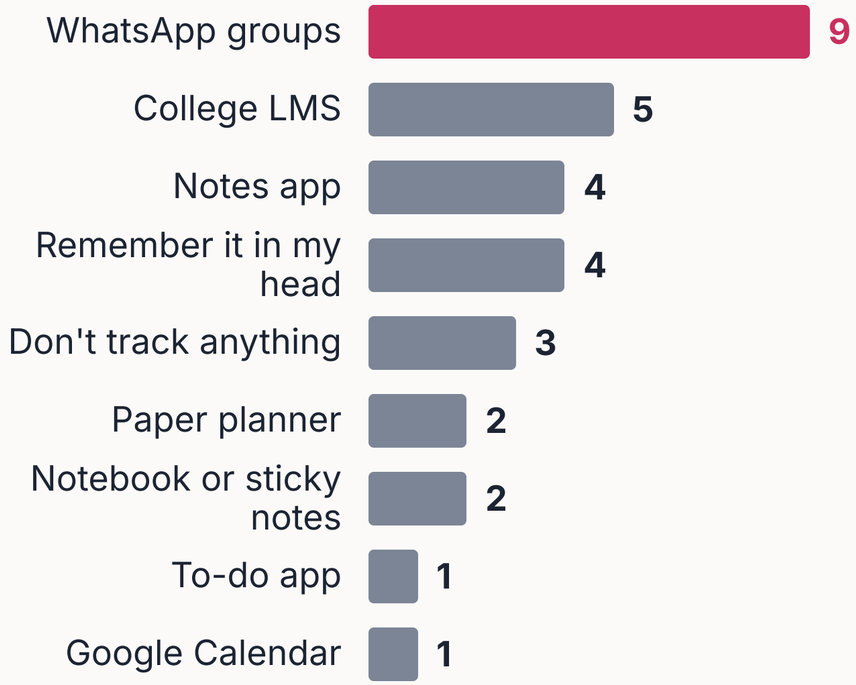
Problems with Moodle

TICK ALL THAT APPLY · n = 11 · HIGHLIGHT = MARKS AND FEEDBACK

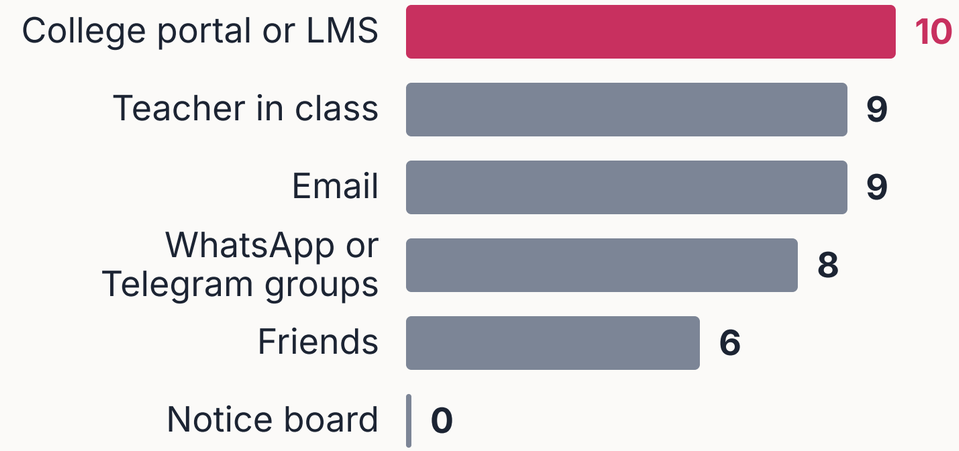


How they track and learn about work

HOW THEY TRACK SCHEDULE AND ASSIGNMENTS



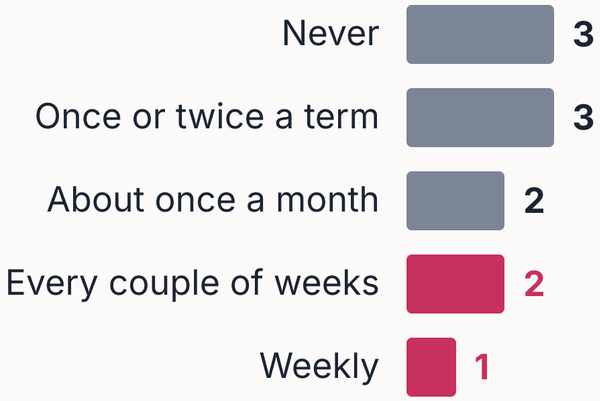
HOW THEY FIND OUT ABOUT DEADLINES



Counts of 11 respondents, tick all.

Deadlines

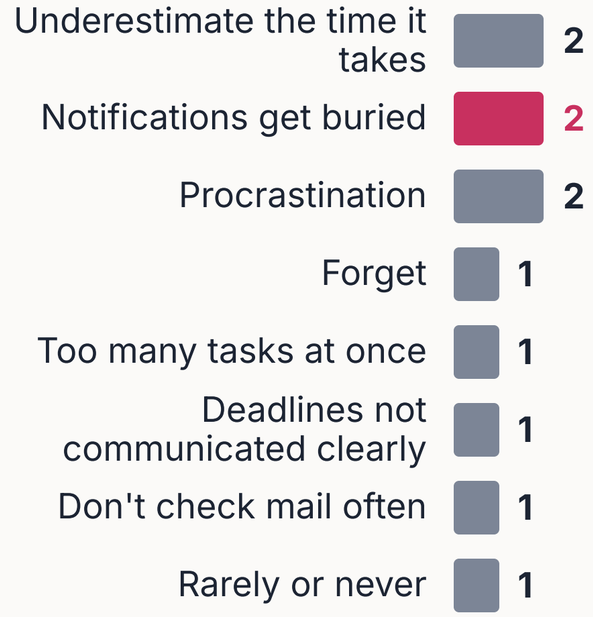
HOW OFTEN THEY MISS OR NEARLY MISS ONE



8 of 11

miss or nearly miss a deadline at least once a term

THE MAIN REASON



Who they are

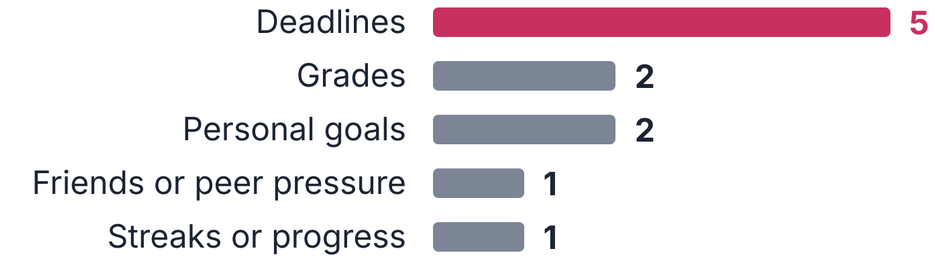
WHICH DESCRIPTION FITS BEST



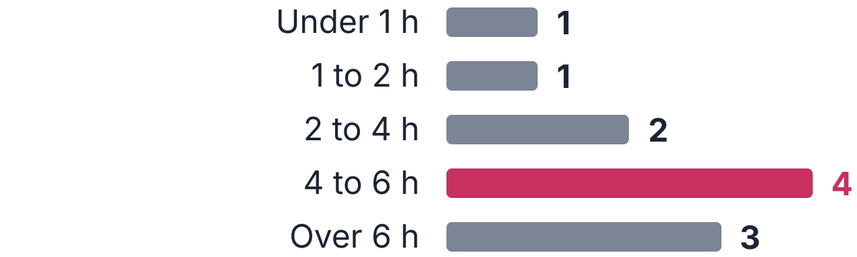
A BIG ASSIGNMENT DUE IN 2 WEEKS



WHAT MOTIVATES MOST



DAILY NON-STUDY SCREEN TIME



n = 11, counts. Year 4: 9, year 3: 2.

How they describe themselves

SEVEN AGREE STATEMENTS · n = 11 · NEUTRAL CENTRED ON THE MIDDLE · NUMBERS ARE RESPONDENTS



Three student personas

The Memoriser — 6 of 15

Knows the week by heart. Watches WhatsApp and mail for changes.

The Daily checker — 8 of 15

Checks the site before class and picks a semester and major every visit.

The Self-organiser — 3 of 15

Copies the week into a calendar or alarms. The copy goes stale.

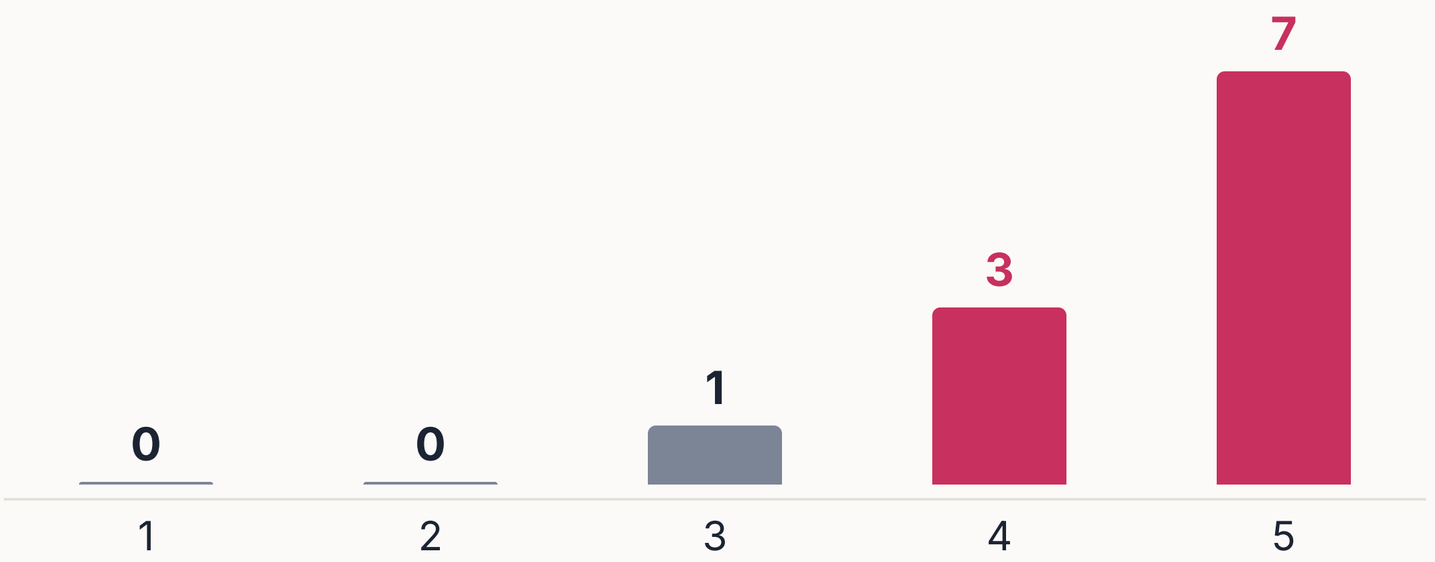
> *"Once, the class mentioned in the mail was different from the one in the timetable, and I only got to know about it later with my friend."*
>
> Student, SICSR

Would they use it?

91%

10 of 11 rate 4 or 5 out of 5 for using an app like this daily

HOW LIKELY TO USE IT DAILY · 1 = NOT AT ALL, 5 = DEFINITELY · n = 11



Faculty: who we met

10
interviewed in September 2026

5 + 5
SICSR and SIG

6 + 4
in person and by phone

1
visiting

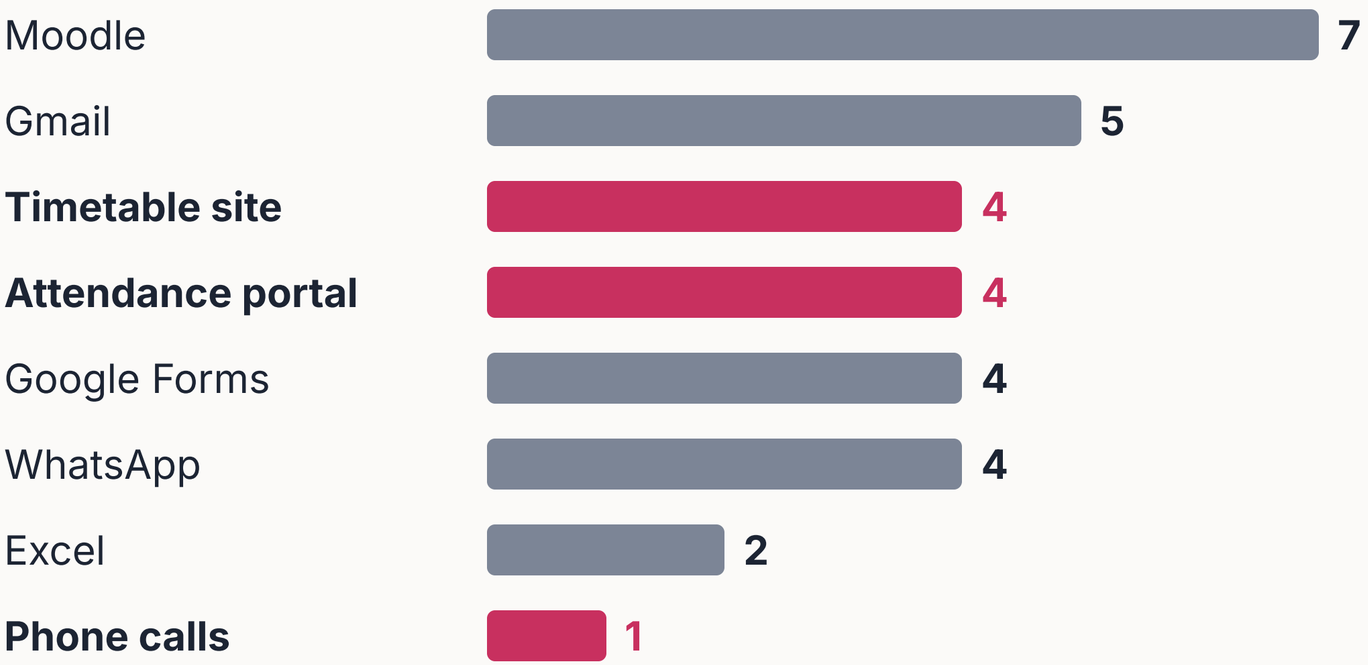
"Every lecture I lose five minutes just setting up attendance."

Faculty, SIG

"Every time I have to explain my whole schedule to the coordinator again."

Visiting faculty, SICSR

Faculty: the tools they juggle



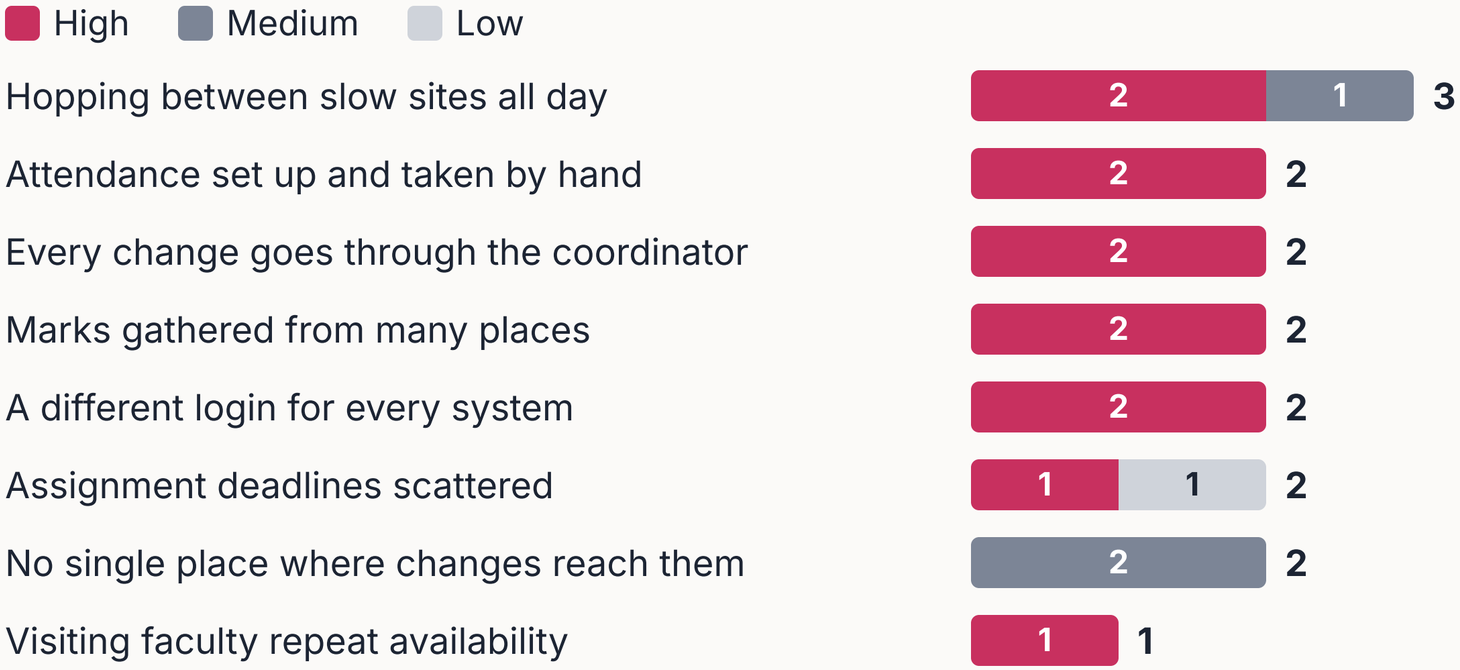
Number of faculty using each tool, n = 10

8

different tools, about 3 each

Highlighted: the timetable site, the attendance portal and phone calls are the ones Plannr takes over.

What faculty raised, and how much it hurts



Faculty who raised each problem, by their own rating. n = 10

Workarounds and what would help

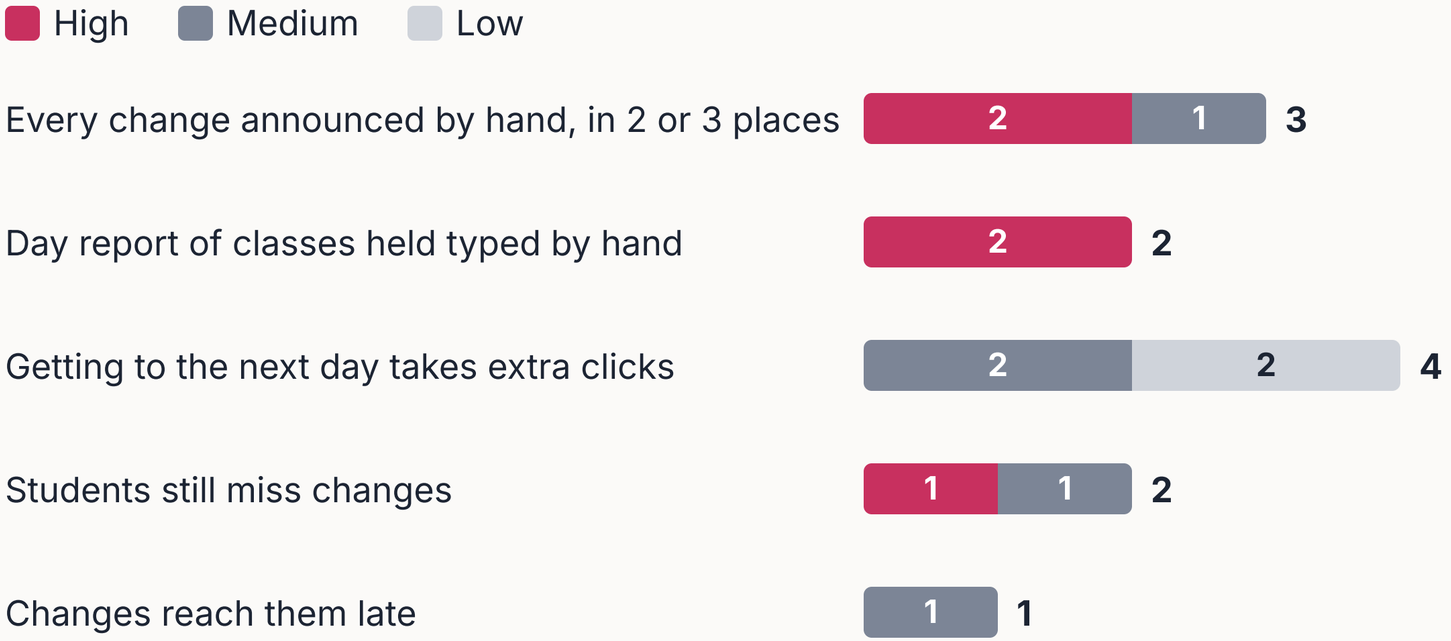
WORKAROUNDS TODAY

- A handwritten timetable pasted on the desk
- A notebook of passwords
- Sticky notes and phone reminders
- Personal Excel sheets
- A second person cross-checks marks
- Attendance in the last minutes of class

WHAT WOULD HELP, IN THEIR WORDS

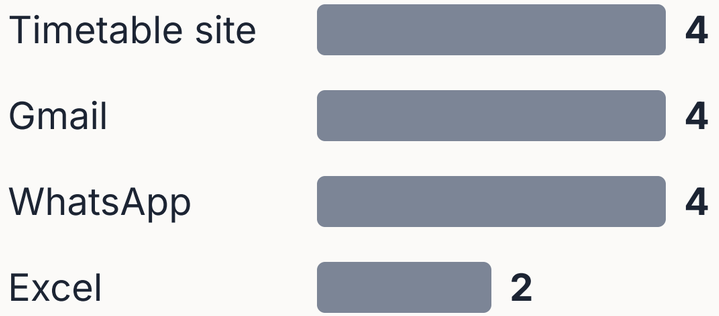
- One login for everything
- A timetable page that opens quickly
- Attendance sessions created automatically from the timetable
- One dashboard of deadlines and notifications
- Entering availability once
- A dashboard where changes are posted once and reach everyone

Coordinators: what they raised

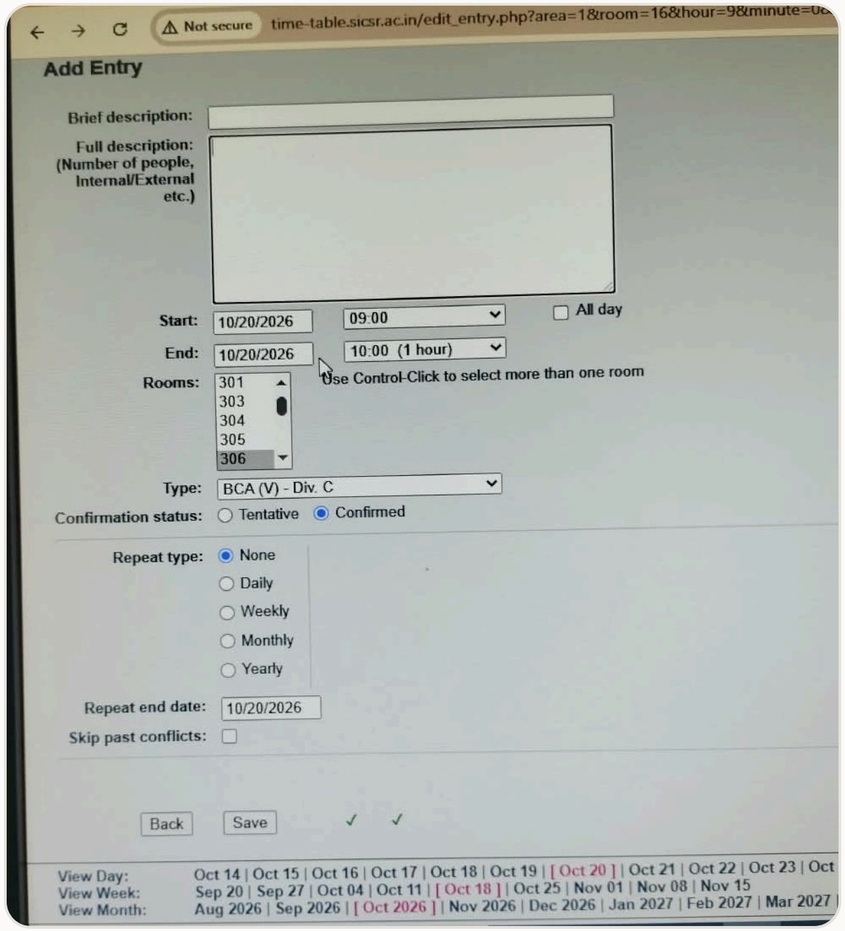


Coordinators who raised each problem, by rating. n = 4

TOOLS THEY USE



The scheduling screen today



WHAT WE OBSERVED

- Each lecture is typed into the MRBS "Add Entry" booking form
- They check Moodle to see how many lectures are left
- They asked for a one-day report and an automated email to all students

Personas across the groups

FACULTY

Internal faculty

Hops between slow sites all day and sets up attendance by hand every lecture.

Visiting faculty

Explains their whole schedule to the coordinator again each time.

COORDINATORS

The relayer

Announces every change by hand, in two or three places.

The record keeper

Enters each lecture by hand and types up the day's report of classes held.

How we prioritised

Score = reach × severity × scope, out of 100

Reach is the share who raised it. Severity and scope are each out of 5.

CORE SCOPE: WEIGHS 5

- Timetable
- Attendance
- Class scheduling and tracking
- Tests
- Assignments

FUTURE SCOPE: WEIGHS 1

- File uploads
- Course material
- Everything else

What ranked highest

STUDENTS

73	Changes not notified
64	Deadlines in different apps
51	Slow, buggy or crashes
49	Cluttered page
49	Too many steps to find classes

FACULTY

24	Hopping between slow sites
20	Attendance by hand
20	Marks gathered by hand
20	Every change via the coordinator

COORDINATORS

60	Every change announced by hand
50	The day's report by hand
40	Getting to the next day
40	Students miss changes

Score out of 100.

What Plannr does

Personal timetable
Only your groups, electives or teaching

Google Calendar sync
Opt-in; moves and cancellations follow

Attendance
A register for every class, from the timetable

Requests and approvals
One queue that shows clashes

Clash detection and room fit
Room, teacher, group, capacity, features

Room booking
Including recurring bookings

Timetable planner
Generates a clash-free week

Recurrence
Stored once as an RRULE

CSV import with preview
Every addition, change, removal

Sign-in and permissions
Google, sicsr.ac.in only

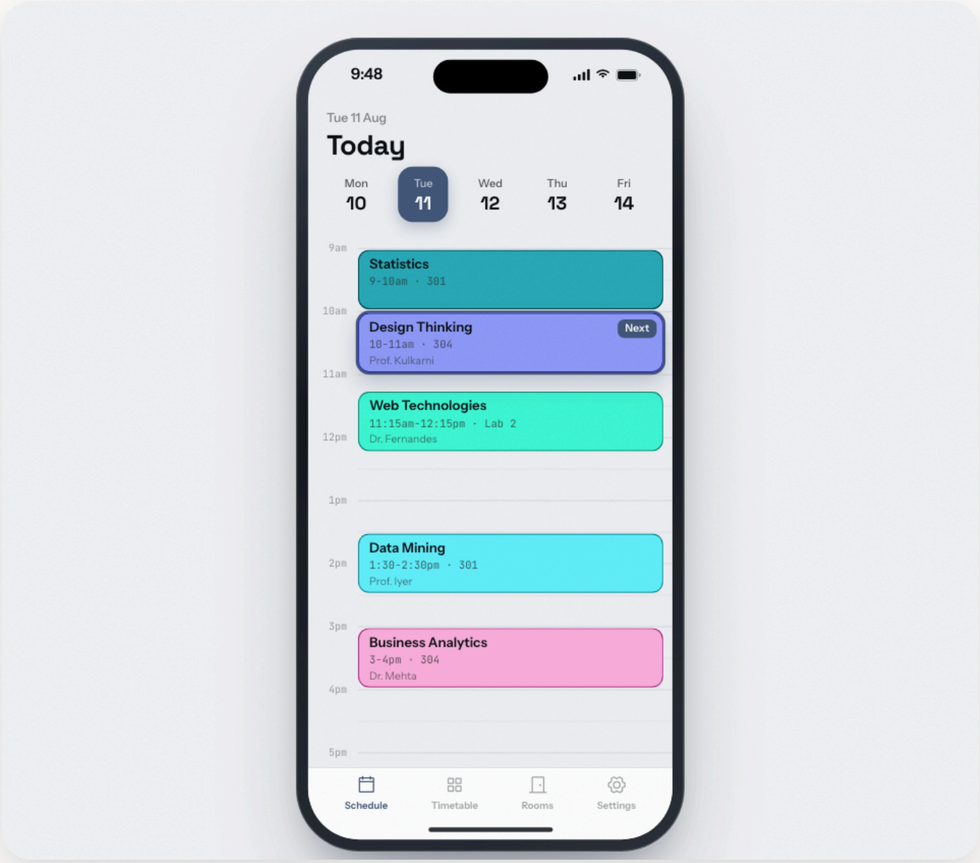
REST API and API keys
OpenAPI 3.1, scoped keys

Siri shortcuts
Via personal API keys

Installable app
Add to the home screen

13 features
All built and live

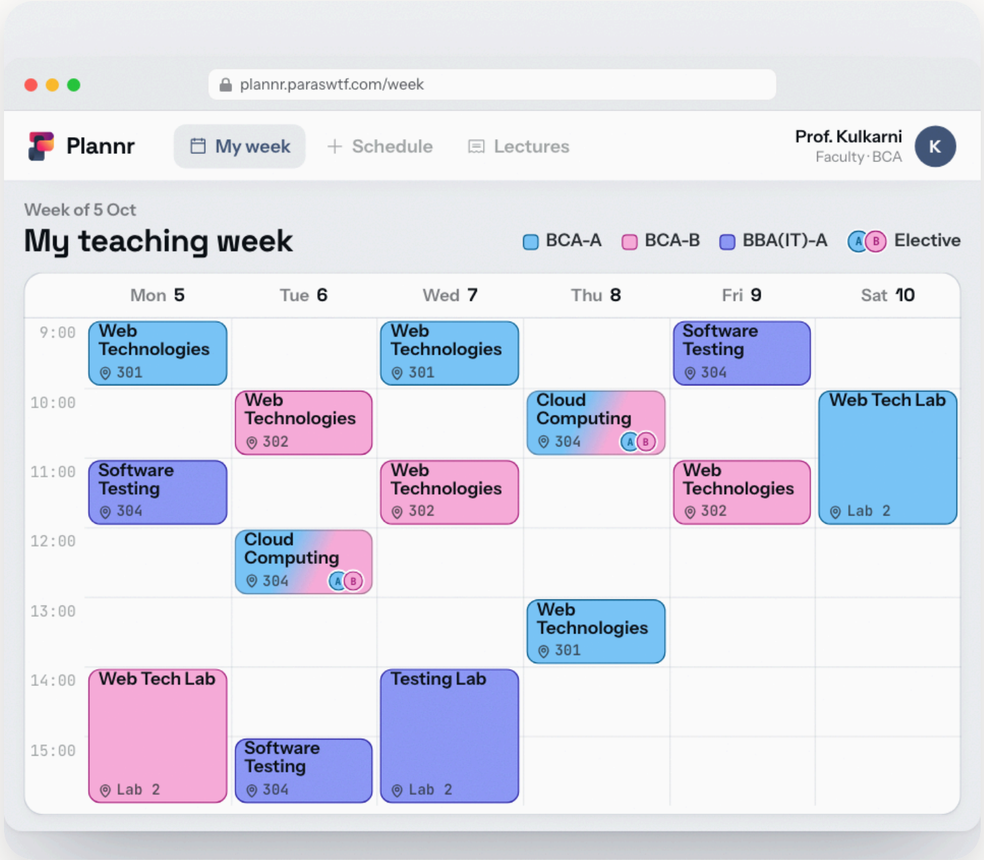
Your own timetable



Plannr's interface

- Only your groups and electives, or only your teaching
- Day, week and month views
- Swipe between days
- Opens on your next class

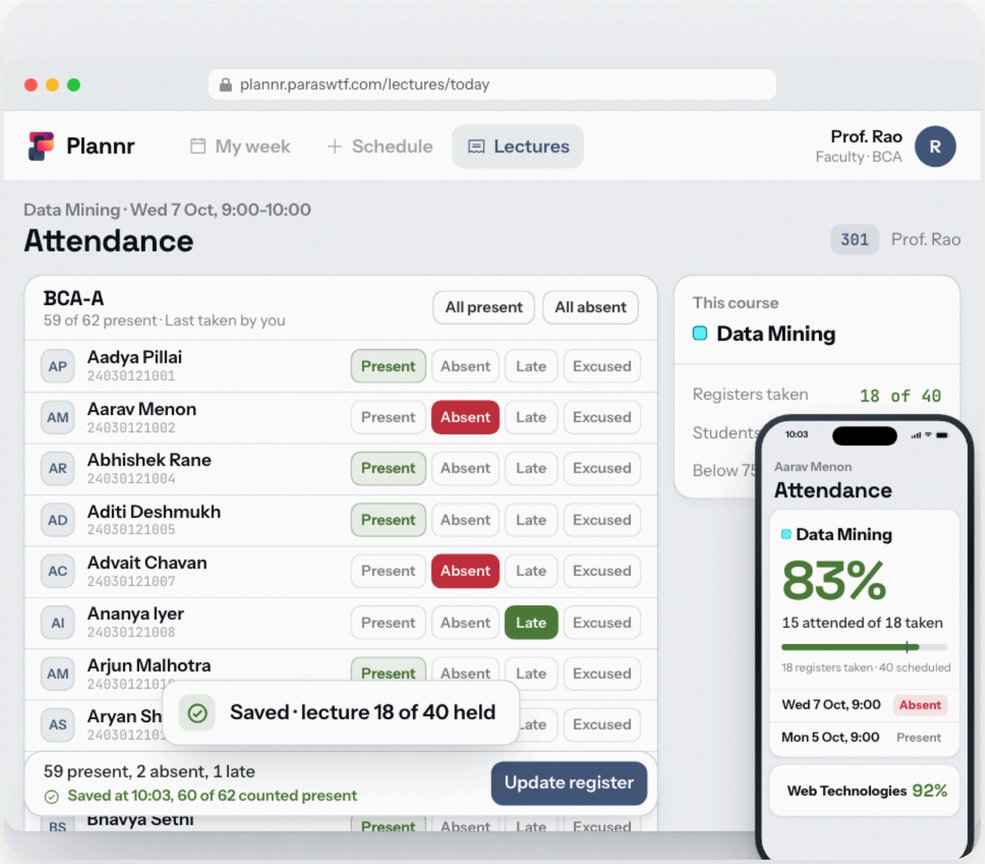
A teacher’s week



Plannr's interface

- Every class you teach in one week
- Combined lectures are shown once
- An edit moves every affected timetable
- Google Calendar sync follows moves and cancellations

Attendance from the timetable

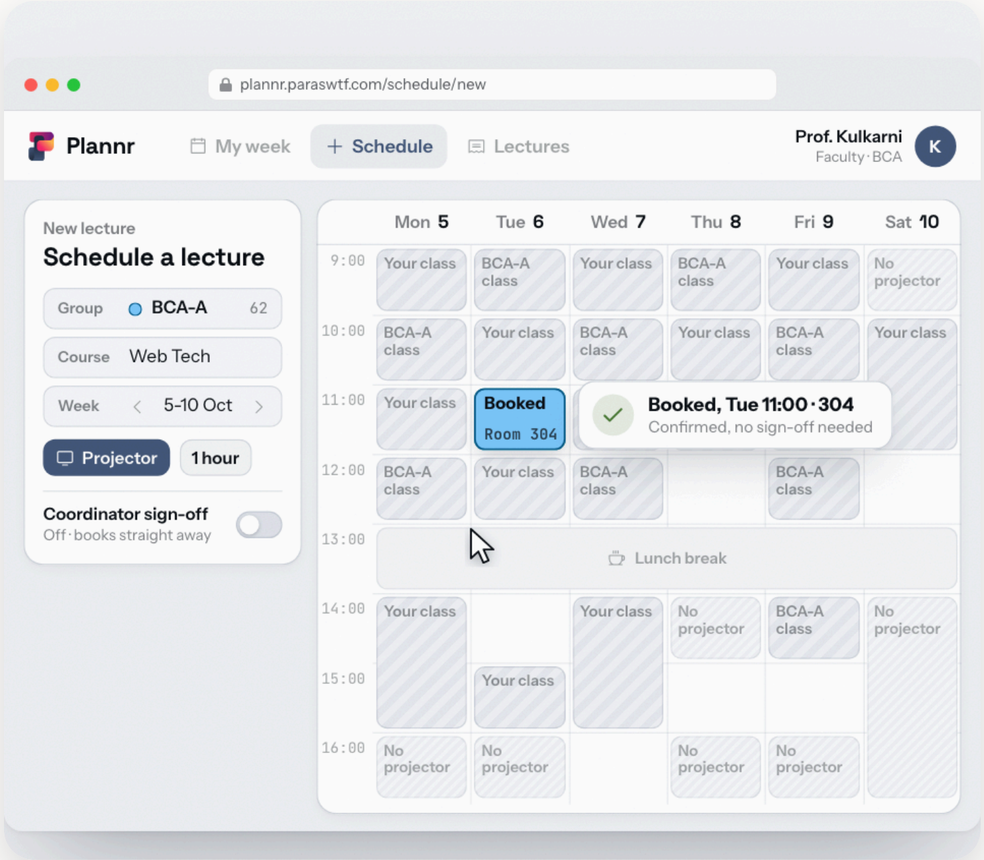


Plannr's interface

- A register exists for every class, nothing to set up
- Mark present, absent, late or excused
- Students see attended, held and remaining per course

Removes the five minutes teachers lose setting up attendance.

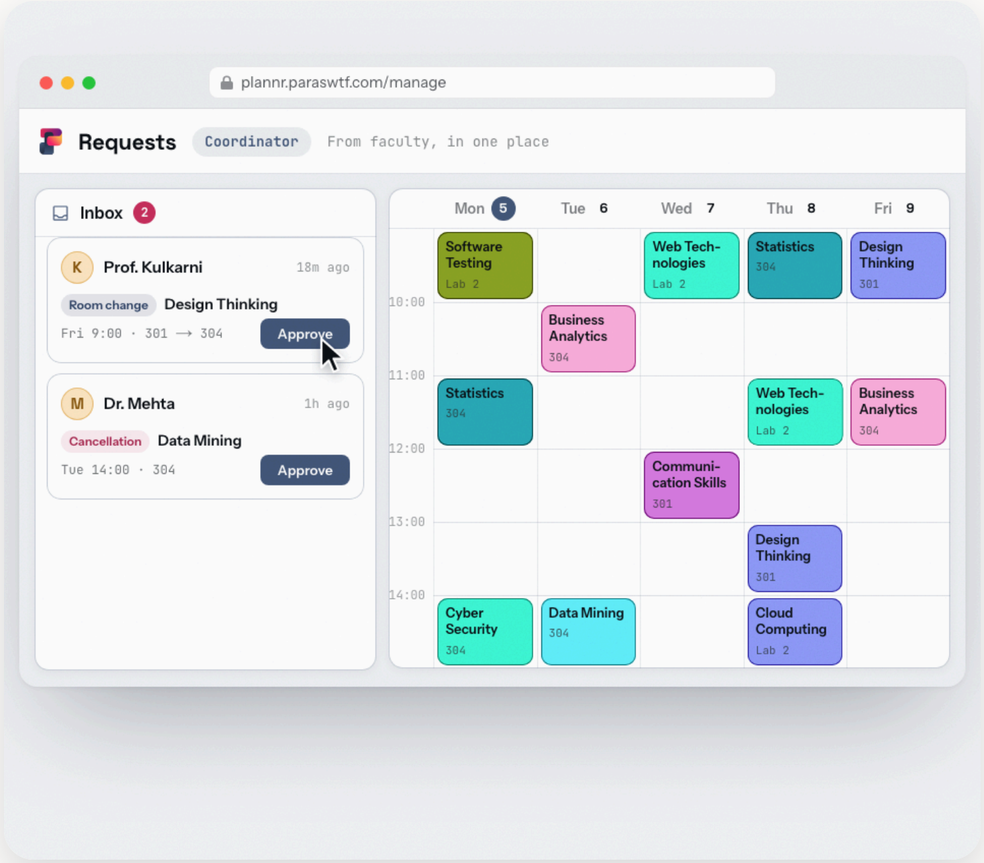
Schedule an extra lecture



Plannr's interface

- Only clash-free slots are offered
- Only rooms that fit the group
- Book it directly, or send a request

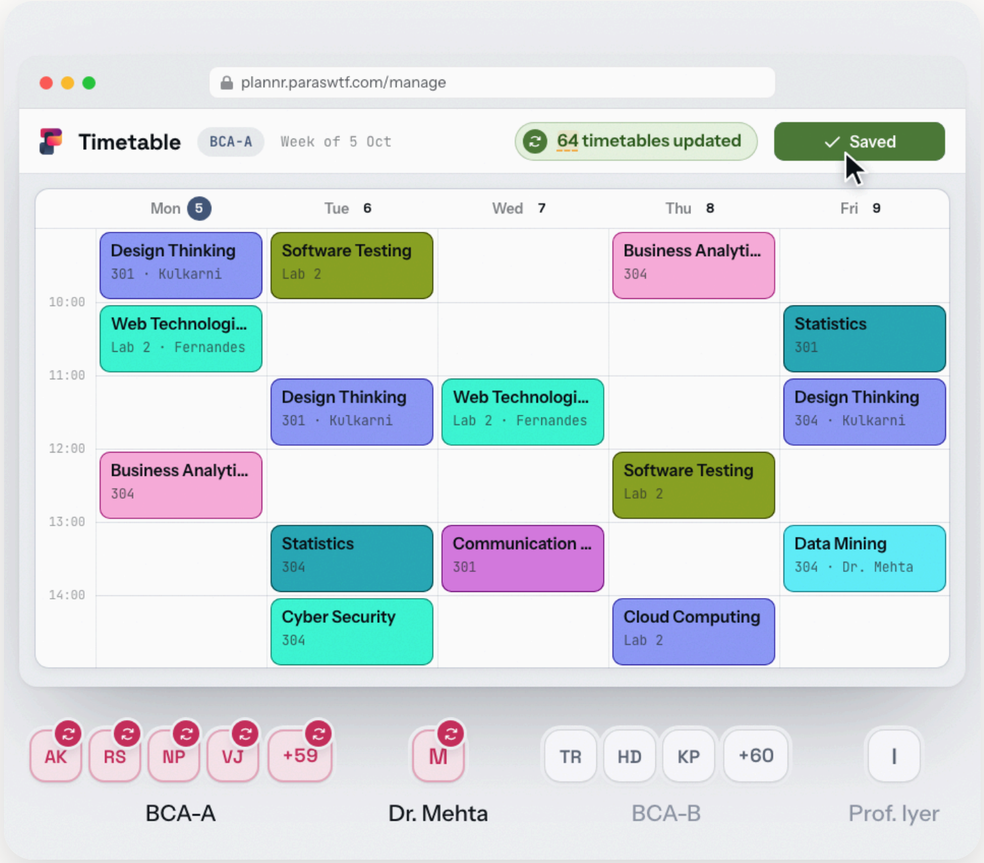
One queue for requests



Plannr's interface

- Extra lectures, room changes and cancellations in one place
- Clashes are shown before you approve
- Approval takes a PostgreSQL advisory lock per room, teacher and group, so two requests cannot get the same slot

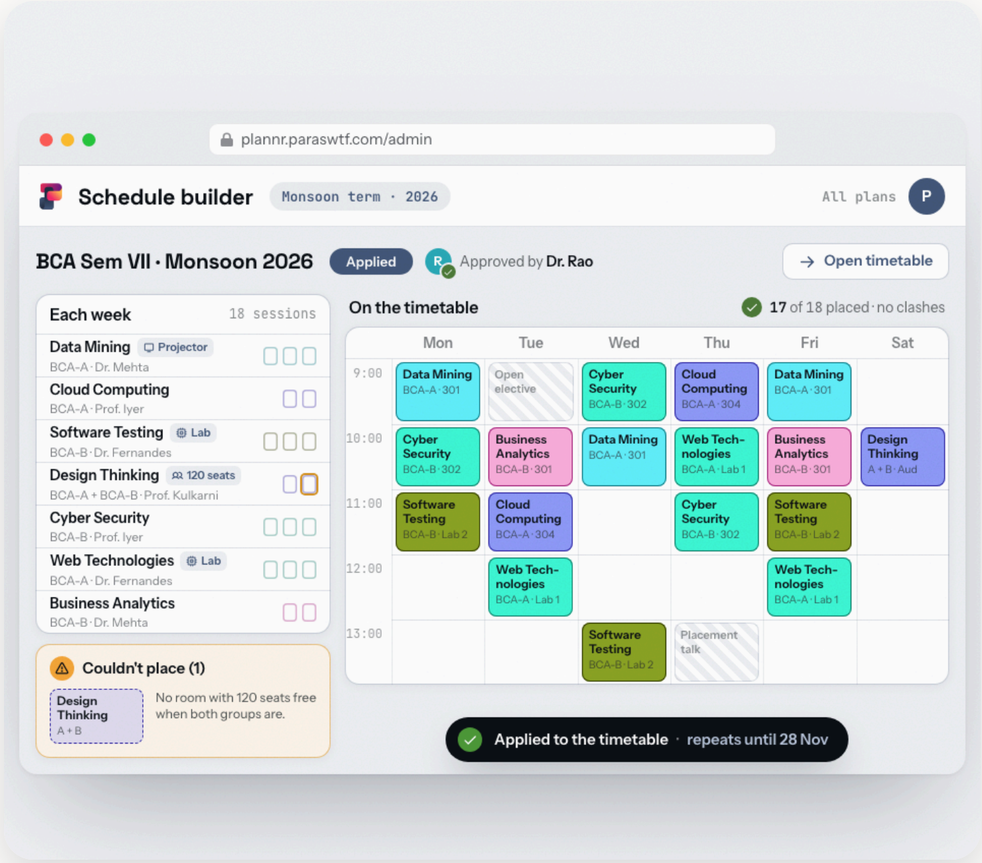
Edit once



Plannr's interface

- A coordinator moves a class once
- Every affected timetable updates
- Students and teachers see the same change

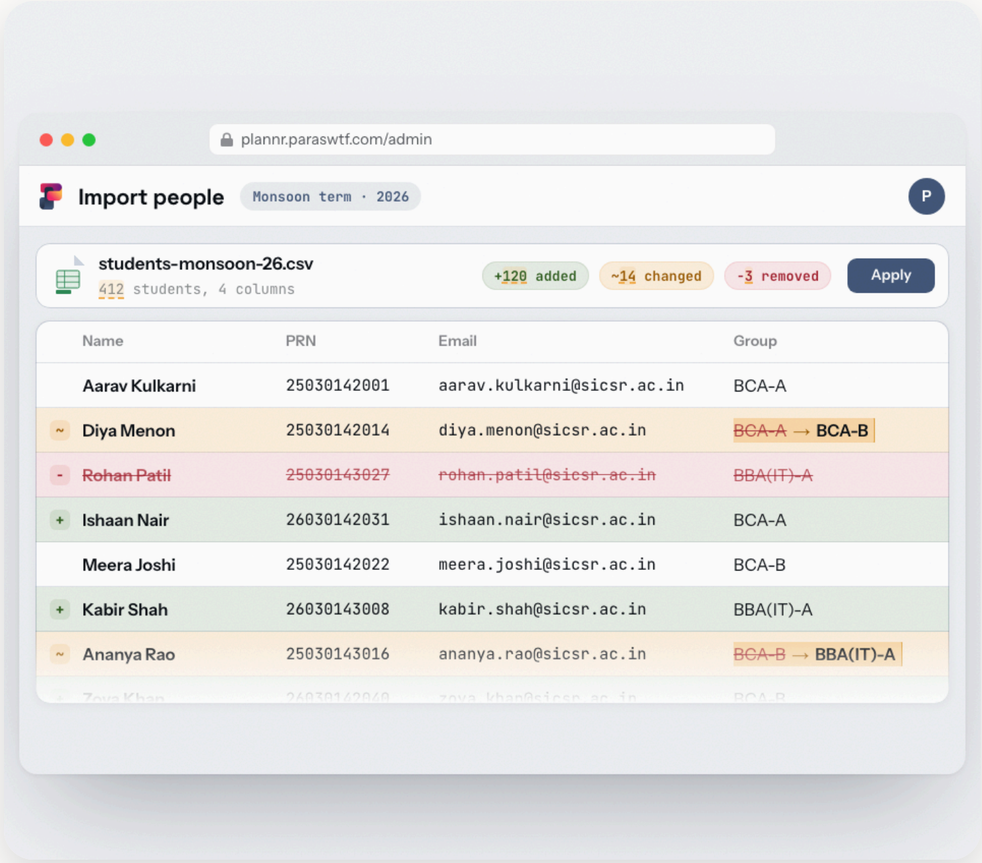
The timetable planner



Plannr's interface

- Weekly demand per group in, a clash-free week out
- Unplaced sessions are listed with reasons
- Review and approval: draft, in review, approved, applied
- Applied to the live timetable as a weekly pattern

Bulk import



Plannr's interface

- Upload a CSV of people or rooms
- Preview every addition, change and removal
- Nothing is saved until you confirm

The platform underneath

College sign-in

Google, sicsr.ac.in only, checked on the server

Roles and permissions

Stored as a bitfield, with role defaults

REST API

/api/v1 with an OpenAPI 3.1 document and scoped keys

Siri shortcuts

Through personal API keys

Installable web app

Add to the home screen

Recurrence

Stored once as an RRULE, materialised in the same transaction

Technology used

Next.js 15 · React 19 · TypeScript · tRPC · Prisma · PostgreSQL · Auth.js + Google · Tailwind CSS v4 · Vercel serverless · REST API + OpenAPI · Google Calendar API · Vitest

SYLLABUS CONCEPTS, AS APPLIED

Ajax
Every screen loads and saves with asynchronous JSON calls (tRPC, REST)

DOM
React updates the DOM; direct DOM APIs for swipe and measuring

Server-side database
Prisma + PostgreSQL in Node.js, the role PHP + MySQL play in LAMP

LAMP integration
Imports and reconciles the college's MRBS timetable, with a preview

Why Next.js, serverless and PostgreSQL

	Plannr's choice	LAMP, MySQL, WordPress
Kind of product	An application with scheduling logic	A CMS would need it all as custom plugins
Language	TypeScript from database to browser	PHP + jQuery splits it in two
Database	PostgreSQL: advisory locks, strict constraints	MySQL: not the locks and constraints Plannr relies on
Hosting	Vercel: HTTPS, preview per change, next to the DB	A server to patch and keep running
Today's proof	The college's LAMP timetable: one 141 KB page, desktop only, plain HTTP	

Built for this course: a legacy version (web-ui-compliance branch) re-creates the timetable viewer on LAMP with jQuery, jQuery UI and Ajax · github.com/paraswtf/plannr/tree/web-ui-compliance

The legacy version on LAMP

web-ui-compliance branch: the timetable viewer re-created with jQuery, jQuery UI and Ajax

SERVER

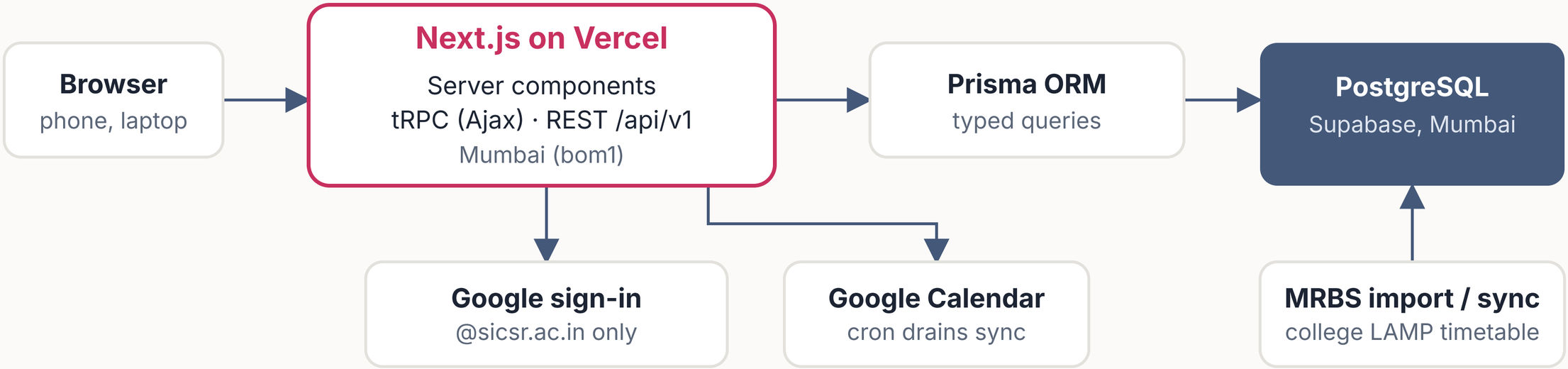
- PHP 8 + PDO endpoints: api/groups.php, day.php, week.php, faculty.php, changes.php
- Clash check on a new change returns 409
- MySQL schema: rooms, groups, faculty, courses, classes, class changes
- Apache .htaccess

BROWSER

- jQuery $.getJSON and $.ajax calls
- DOM rendering of the day and week
- jQuery UI datepicker, dialog and autocomplete
- Changes feed polled every 30 s

Code: **https://github.com/paraswtf/plannr/tree/web-ui-compliance/web-ui-compliance**

System architecture



REQUEST FLOW



Workflow: from request to timetable

1

Request

Faculty asks for an extra lecture, a room change or a cancellation

→

2

Clash check

Room, teacher, group and capacity; next free slot

→

3

Approval

Coordinator approves; advisory lock per room, teacher, group

→

4

Occurrences

Regenerated in one transaction

→

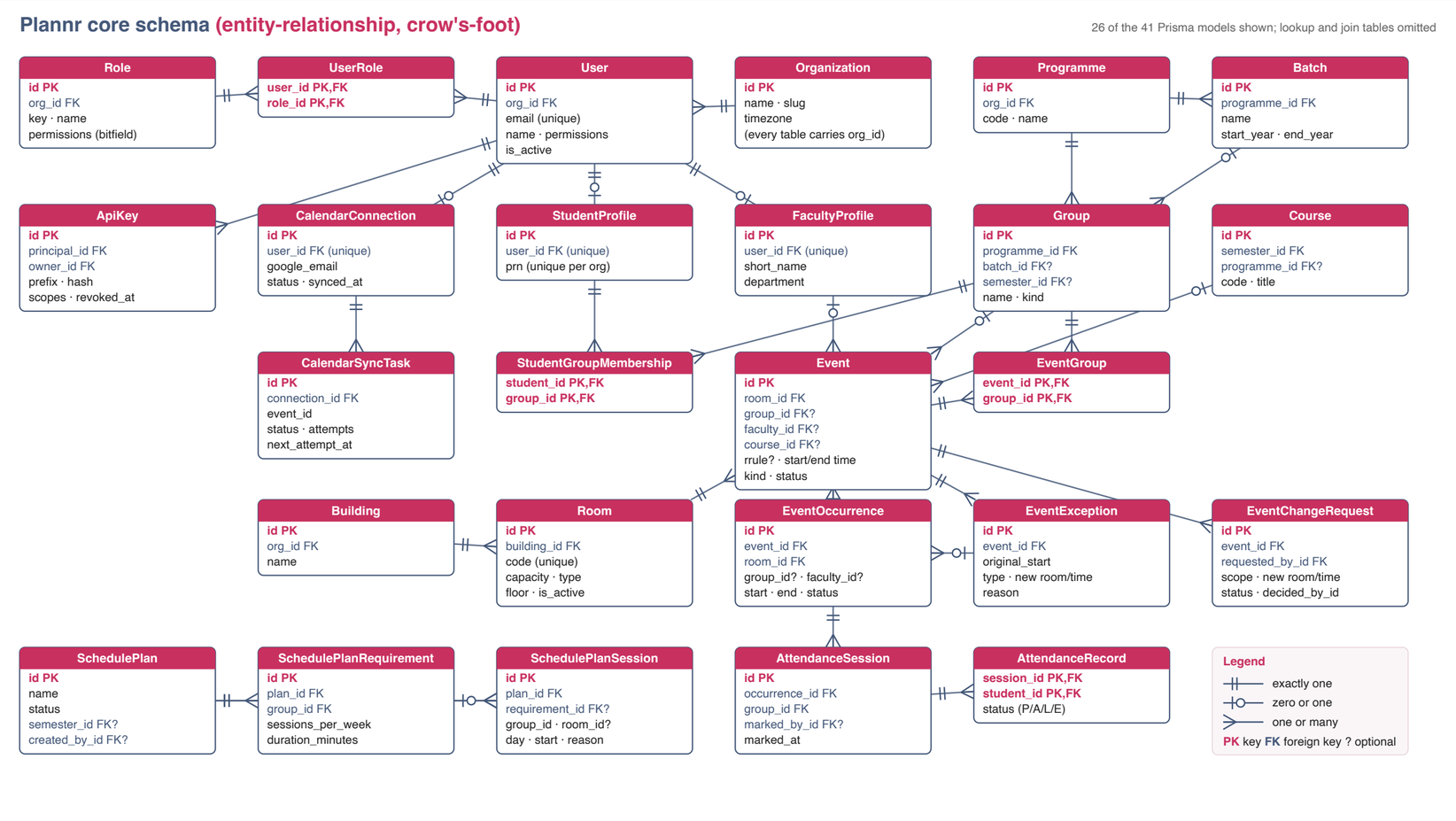
5

Follow

Calendars and timetables update

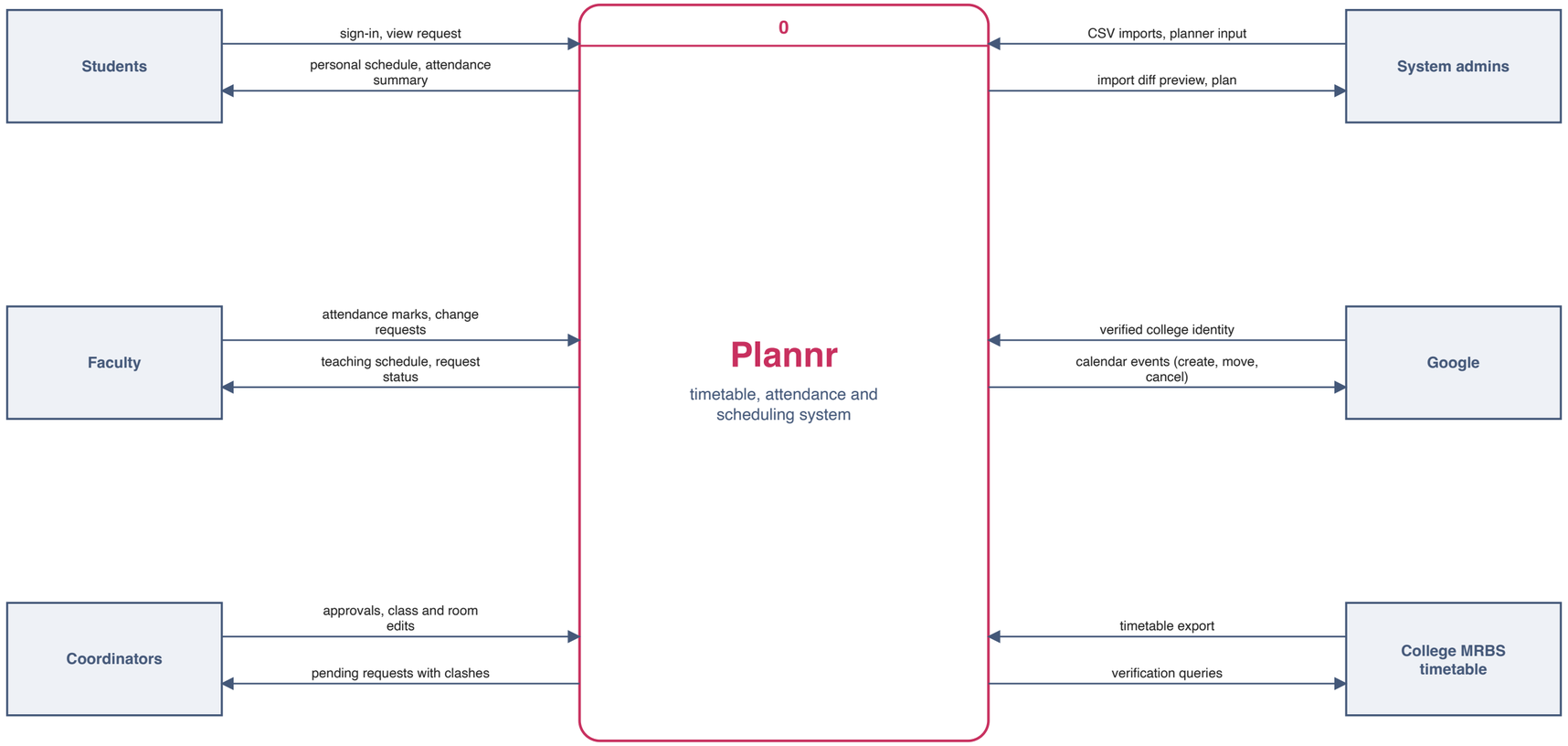
The lock means two requests cannot both get the same slot. Students with Google Calendar sync see moves and cancellations follow; cancelled classes are retitled, not deleted.

Data design: ER diagram



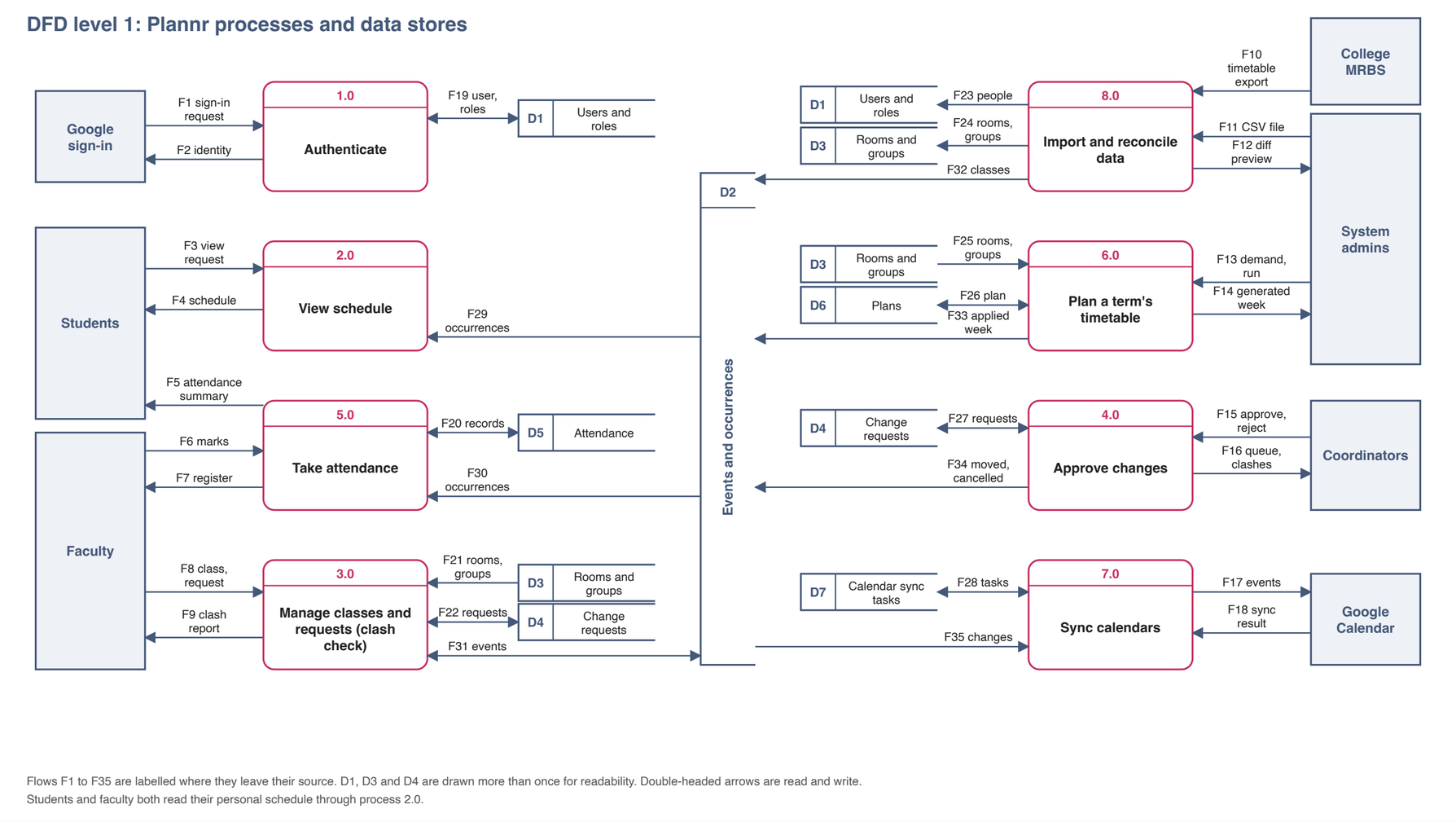
Data flow: DFD level 0

DFD level 0: Plannr context diagram



Arrows show data flowing in the direction of the head. Google appears once here; sign-in and Calendar sync are two separate flows.

Data flow: DFD level 1



Flows F1 to F35 are labelled where they leave their source. D1, D3 and D4 are drawn more than once for readability. Double-headed arrows are read and write. Students and faculty both read their personal schedule through process 2.0.

Testing

49

Vitest test files

Suite	Covers
Unit tests	Recurrence, conflicts, free slots, permissions, CSV diff, the planner's solver, pagination, OpenAPI, crypto, timetable diff, schedule views, dates
Integration tests	Against a local PostgreSQL: timetable, events, approvals, attendance, courses, programmes, the REST handler, Google Calendar sync, personal API keys
db:verify-schedule	Compares the database with the live college timetable, class by class

Demo path

1. Sign in
2. My day
3. A teacher's week
4. Take attendance
5. Request an extra lecture
6. Approve it
7. The planner
8. Import

Live: **plannr.paraswtf.com**
Code: **github.com/paraswtf/plannr**
Legacy version: **github.com/paraswtf/plannr/tree/web-ui-compliance/web-ui-compliance**

Individual contribution

Each member researched their stakeholders and developed those stakeholders' features.

Member	Contribution
Paras Deshpande	Research with system admins; architecture; developed the admin features: CSV import, timetable planner, MRBS sync
Sonal Mohrir	Research with students; developed the student features: personal timetable, Google Calendar sync
Tanisha Nath	Research with faculty and coordinators; developed their features: attendance register, lecture requests, approvals queue, clash checks, room booking

Conclusion

- Plannr turns the college's single desktop page into a personal timetable on any device
- Attendance, requests and approvals run in one place, with clash checks and locks
- It builds on the data the college already holds, imported from MRBS with a preview
- A legacy version applies the syllabus stack: PHP, MySQL, Apache, jQuery and Ajax

Future scope

01 Push and email change notifications

02 The day's report of classes held

03 Faculty availability entry

04 Check-in codes

05 Tests and marks

06 Assignments and deadlines

07 Course material uploads

08 An admin dashboard

09 Offline mode

Thank you

Questions?

plannr.paraswtf.com

github.com/paraswtf/plannr